

QUERY-AWARE PHRASE-TO-REGION ALIGNMENT FOR TEXT-BASED PERSON SEARCH

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What?



Text-Based Person Search aims to find target person images from a given natural language description.



It requires joint understanding of textual descriptions and visual information about people.



The task is evaluated by ranking results based on relevance to the query.

Why?



Person search is in high demand for security monitoring, helping to quickly identify and locate wanted individuals.



This task bridges research in computer vision, natural language processing, and information retrieval.



The approach helps improve the alignment between text descriptions and visual regions, leading to more accurate and efficient retrieval results.

Overview

Text token Image token / Patch token Global text token (T_{EOS}) Global image token (T_{CLS}) Transport plan

Batch text

A man wearing a blue jacket, black pants and carrying a black backpack.



Text tokens



T_{EOS}

Batch images



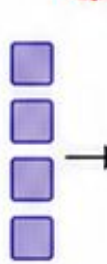
Image tokens



T_{CLS}

Global contrastive loss

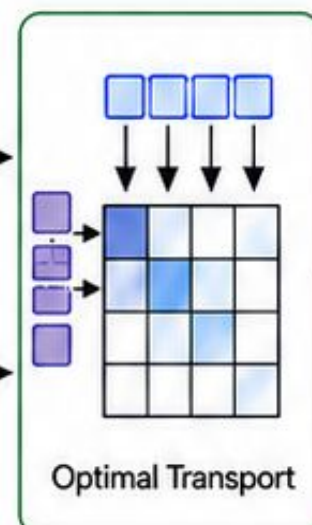
Query-Aware Alignment



Top- K_t



Top- K_r



Similarity

0.82

0.35

0.21

...

0.07

1. Problem Statement



Input:

Text query and a set of person images.



Output:

Rank images according to relevance to the query.



Challenge:

- Ambiguity in language and vision.
- Variations in pose, lighting and background.
- Need for precise cross-modal alignment.

2. Proposed Method

A. Clip Baseline

- Use image encoder and text encoder to extract global features.
- Calculate global image and global text embeddings.



B. Query-Aware Token Selection

- Select important text and image tokens.
- Retain tokens that best capture the query meaning.



C. Phrase-to-Region Alignment

- Find matching regions between text phrases and image patches.
- Optimize the transport matrix to use Optimal Transport.
- Align regions to improve text descriptions using image patches.



3. Evaluation



Datasets:

- CUHK-PEDES
- ICFG-PEDES
- RSTReid



Metrics:

- Rank-1, Rank-5, Rank-10
- mAP



Additional goal:

- Visualize and analyze alignment to improve understanding of the method.

Expected Results



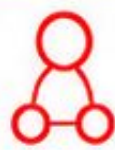
Framework

Build a unified framework for person retrieval with strong text-image alignment.



Effectiveness

Improve retrieval accuracy through better query understanding.



Local Alignment

Achieve better fine-grained alignment between text phrases and image regions.



Performance

Outperform current state-of-the-art methods on Text-Based Person Search benchmarks.